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Estimation of the acentric factor and the liquid molar volume at 298 K using a new group contribution method

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Abstract

The method of Constantinou and Gani [Constantinou, L. and Gani, R., 1994. A new group contribution method for the estimation of properties of pure compounds. AIChE J., 40(10): 1697–1710] is extended to the estimation of the acentric factor and the liquid molar volume at 298 K. Thus, it now allows reliable prediction of all the common corresponding-state physical properties. In addition to the accurate estimation of liquid molar volumes at 298 K for low-molecular-weight compounds, the technique can be used for amorphous polymers.

Keywords: Theory; Application; Group contribution; Liquid molar volume; Acentric factor; Pure compounds

1. Introduction

Process synthesis and simulation as well as product design require information on the properties of pure compounds and mixtures. The needs have always been greater than the available experimental data, leading to the development of many property estimation methods. Generally, the simplest successful methods are based on the concept of additive group contributions, but many of these are limited in application and reliability.

Recently, Constantinou and Gani (1994) developed a new group contribution method for the estimation of thermodynamic and physical properties of pure compounds. The method uses information on the molecular structure of a compound to estimate its properties at two levels. The basic level has contributions from first-order functional groups, such as those applied in

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many previous methods (Reid at al., 1987). The second level uses a set of second-order groups, which distinguish alternative collections of first-order groups as suggested by the conjugation concept (Constantinou and Gani, 1994). In this way, the method allows for both a first-order estimation and a more accurate second-order approximation.

The method was successfully applied to the estimation of eight physical and thermodynamic properties of pure compounds: normal boiling point, normal melting point, critical pressure, critical temperature, critical volume, standard enthalpy of vaporization at 298 K, standard Gibbs energy and standard enthalpy of formation at 298 K. The results demonstrated improved accuracy in all cases, distinctions were made among isomers, and reliable extrapolation with alkane chain length was obtained for the critical temperature, critical pressure and normal boiling point.

In this work, we demonstrate further applicability of the Constantinou and Gani (1994) method for prediction of the acentric factor and the liquid molar volume at 298 K, including prediction of the liquid density of amorphous polymers using contributions determined only from smaller substances.

2. Estimation of the acentric factor

The acentric factor of Pitzer (1955) has become one of the most important properties used in corresponding states calculations of pure compounds and mixtures. The definition of the acentric factor by Pitzer (1955) characterizes the non-sphericity of molecular interactions:

$$w = -1 - \log_{10}(P_r^{sat})_{T_r = 0.7} \tag{1}$$

The value of the acentric factor is zero for noble gases and greater than zero for polyatomic substances. It was not intended to take into account polar and association/solution effects, although it has often been used successfully for such substances when other parameters are included. When the acentric factor cannot be estimated from its definition (Eq. 1), property estimation techniques have to be used. These methods originate from two different bases. The first one uses other primary properties. Thus, Edmister (1958) and Lee and Kesler (1975) require the normal boiling point, the critical temperature, and the critical pressure, while Lin and Chao (1984) use molar mass, normal boiling point, and relative density. The need for other primary properties significantly limits the applicability and reliability of such techniques, as experimental values of these properties may not be available. The other estimation methods use group contributions directly. Thus, Hoshino et al. (1982) presented a method for alkanes using only the molecular structure. Han and Peng (1993) proposed another group-contribution technique for both hydrocarbons and non-hydrocarbons. For the latter method, in addition to primary group contributions, the Platt number (the total number of pairs of carbon atoms three bonds apart) must be used for the estimation. One complication is that the contributions for some groups depend on the type of compound in which they appear, making automatic computation difficult.

Given these limitations, and recognizing that being able to treat all corresponding-state properties in the same way is a powerful tool in product and process design, we extend the model of Constantinou and Gani (1994) to the acentric factor. The basic equation is

$$\exp(w/a)^b - C = \sum_i N_i w_{1i} + A \sum_j M_j w_{2j} \tag{2}$$

where w_{1i} is the contribution of the first-order group type i, which occurs N_i times, and w_{2j} is the contribution of the second-order group type j, with M_j occurrences in a compound. The left-hand-side of Eq. (2) was chosen to provide the best possible fit of the experimental data, to achieve optimal additivity of the group contributions and to provide reliable extrapolation behavior (Constantinou and Gani, 1994). a, b, and c are universal quantities. The constant A is used to distinguish the second-level estimation (where both first- and second-order group contributions are involved, so $A = 1$) from the first-level estimation (only a first-order approximation, so $A = 0$). The determination of the first- and second-order groups is described in detail by Constantinou and Gani (1994) and no new groups have been added here.

The "data" used in the regression were determined using Eq. (1), for which the experimental values of the critical temperature, critical pressure, and vapor pressure at $T = 0.7Tc$ were taken from the DIPPR-databank (Daubert and Danner, 1989). The estimated uncertainties of these values range between 2 and 4%, which heavily influences the prediction of the acentric factor (Reid et al., 1987). Through regression, the contributions of the first- and second-order groups of Constantinou and Gani (1994) were determined and are given in Tables 1 and 2. For some groups, sufficient data could not be found to determine their contributions (designated by **** in the Tables). The universal parameters a, b, and c have the values 0.4085, 0.5050, 1.1507, respectively. Examples of how the method can be used are given in Appendix A. Statistics of the regression are given in Table 3. The average absolute percentage error (AAPE) is 3.0% (which is likely to be within experimental uncertainty), as opposed to the value of 4.2% reported by Han and Peng (1993).

3. Liquid molar volume

The saturated liquid molar volume at 298 K is estimated. Existing methods for the estimation of this property can also be divided into two categories: those that are functions of critical properties, acentric factor, normal boiling point, and other parameters characteristic to each compound, such as the Rackett compressibility factor (Rackett, 1970; Spencer and Danner, 1972; Spencer and Adler, 1978; Hankinson and Thomson, 1979); simple group-contribution techniques (Elbro et al., 1991; Fedors, 1974). While the first category is applicable only to low-molecular-weight compounds, the latter has also been applied to polymers. In all cases, the models are specific to the liquid molar volume and provide only a first-order approximation with no distinction among isomers. For the liquid molar volume at 298 K, the model of Constantinou and Gani (1994) takes the form

$$V_1 - d = \sum_i N_i v_{1i} + A \sum_j M_j v_{2j} \tag{3}$$

where v_{1i} is the contribution of the first-order group type i, which occurs N_i times, v_{2j} is the contribution of the second-order group type j, with M_j occurrences in a compound, and d is a universal constant. The experimental data were taken from the DIPPR-databank. Contributions of the first- and second-order groups determined after regression are presented in Tables 1 and 2. The value of d is 0.01211 m^3 $kmol^{-1}$. Statistics of the regression are given in Table 3. It can be noted that the Elbro et al. (1991) method, which is not applicable to classes such as

Table 1
First-order groups and their contributions for estimation of acentric factors and liquid molar volumes at 298 K

Group	w_{1i}	v_{1i} (m^3 $kmol^{-1}$)
CH_3	0.29602	0.02614
CH_2	0.14691	0.01641
CH	−0.07063	0.00711
C	−0.35125	−0.00380
CH_2=CH	0.40842	0.03727
CH=CH	0.25224	0.02692
CH_2=C	0.22309	0.02697
CH=C	0.23492	0.01610
C=C	−0.21017	0.00296
CH_2=C=CH	0.73865	0.04340
ACH	0.15188	0.01317
AC	0.02725	0.00440
$ACCH_3$	0.33409	0.02888
$ACCH_2$	0.14598	0.01916
ACCH	−0.08807	0.00993
OH	1.52370	0.00551
ACOH	0.73657	0.01133
CH_3CO	1.01522	0.03655
CH_2CO	0.63264	0.02816
CHO	0.96265	0.02002
CH_3COO	1.13257	0.04500
CH_2COO	0.75574	0.03567
HCOO	0.76454	0.02667
CH_3O	0.52646	0.03274
CH_2O	0.44184	0.02311
CH–O	0.21808	0.01799
FCH_2O	0.50922	0.02059
CH_2NH_2	0.79963	0.02646
$CHNH_2$	****	0.01952
CH_3NH	0.95344	0.02674
CH_2NH	0.55018	0.02318
CHNH	0.38623	0.01813
CH_3N	0.38447	0.01913
CH_2N	0.07508	0.01683
$ACNH_2$	0.79337	0.01365
C_5H_4N	****	0.06082
C_5H_3N	****	0.05238
CH_2CN	****	0.03313
COOH	1.67037	0.02232
CH_2Cl	0.57021	0.03371
CHCl	****	0.02663
CCl	****	0.02020
$CHCl_2$	0.71592	0.04682
CCl_2	****	****
CCl_3	0.61662	0.06202
ACCl	****	0.02414

Table 1 (continued)

Group	w_{1i}	v_{1i} (m^3 $kmol^{-1}$)
CH_2NO_2	****	0.03375
$CHNO_2$	****	0.02620
$ACNO_2$	****	0.02505
CH_2SH	****	0.03446
I	0.23323	0.02791
Br	0.27778	0.02143
CH≡C	0.61802	****
C≡C	****	0.01451
Cl–(C=C)	****	0.01533
ACF	0.26254	0.01727
$HCON(CH_2)_2$	****	****
CF_3	0.50023	****
CF_2	****	****
CF	****	****
COO	****	0.01917
CCl_2F	0.50260	0.05384
HCClF	****	****
$CClF_2$	0.54685	0.05383
F (except as above)	0.43796	****
$CONH_2$	****	****
$CONHCH_3$	****	****
$CONHCH_2$	****	****
$CON(CH_3)_2$	****	0.05477
$CONCH_3CH_2$	****	****
$CON(CH_2)_2$	****	****
$C_2H_5O_2$	****	0.04104
$C_2H_4O_2$	****	****
CH_3S	****	0.03484
CH_2S	0.42753	0.02732
CHS	****	****
C_4H_3S	****	****
C_4H_2S	****	****

cyclohydrocarbons and organic compounds with nitrogen or sulphur atoms, reports an approximately 1% AAPE for the liquid molar volume at 298 K, while our results give an AAPE of 0.89% for all classes of compounds. (Fedors, 1974, does not provide any information on the accuracy of his method.)

4. Discussion

Table 4 presents the AAPEs for both the estimated properties in terms of classes of compounds: (1) hydrocarbons (C, H); (2) alcohols, aldeydes, ketones, acids and esters (C, H, O); (3) amides

Table 2
Second-order groups and their contributions for estimation of acentric factors and liquid molar volumes at 298 K

Groups	w_{2j}	v_{2j} (m^3 $kmol^{-1}$)	Sample assignments (occurrences)
$(CH_3)_2CH$	0.01740	0.00133	2-Methylpentane (**1**)
$(CH_3)_3C$	0.01922	0.00179	2,2,Dimethylpentane (**1**), 2,2,4,4-tetramethylpentane (**2**)
$CH(CH_3)CH(CH_3)$	−0.00475	−0.00203	2,3-Dimethylpentane (**1**), 2,3,4-tetramethylpentane (**2**)
$CH(CH_3)C(CH_3)_2$	−0.02883	−0.00243	2,2,3-Dimethylpentane (**1**), 2,2,3,4,4-pentamethylpentane (**2**)
$C(CH_3)_2C(CH_3)_2$	−0.08632	−0.00744	2,2,3,3-Dimethylpentane (**1**), 2,2,3,3,4,4-hexamethylpentane (**2**)
3-Membered ring[a]	0.17563	****	Cyclopropane (**1**)
4-Membered ring[a]	0.22216	****	Cyclobutane (**1**)
5-Membered ring[a]	0.16284	0.00213	Cyclopentane (**1**), ethylcyclopentane (**1**)
6-Membered ring[a]	−0.03065	0.00063	Cyclohexane (**1**), methylcyclohexane (**1**)
7-Membered ring[a]	−0.02094	−0.00519	Cycloheptane (**1**), ethylcycloheptane (**1**)
$CH_n{=}CH_m{-}C_p{=}C_k$ $k,n,m,p \in (0, 2)$	0.01648	−0.00188	1,3-Butadiene (**1**)
$CH_3{-}CH_m{=}CH_n$ $m,n \in (0, 2)$	0.00619	0.00009	2-Butene (**2**), 2-methyl-2-butene (**3**)
$CH_2{-}CH_m{=}CH_n$ $m,n \in (0, 2)$	−0.0115	0.00012	1,4-Pentadiene (**2**)
$CH{-}CH_m{=}CH_n$ or $C{-}CH_m{=}CH_n$ $m,n \in (0, 2)$	0.02778	0.00142	4-Methyl-2-pentene (**1**)
Alicyclic side chain $C_{cyclic}C_m$ $m > 1$	−0.11024	−0.00107	Ethylcyclohexane (**1**), propylcycloheptane (**1**)
CH_3CH_3	−0.1124	****	Ethane (only)
CHCHO or CCHO	****	−0.00009	2-Methyl butanaldeyde (**1**)
CH_3COCH_2	−0.20789	−0.00030	2-Pentanone (**1**)
CH_3COCH or CH_3COC	−0.16571	−0.00108	3-Methyl-2-pentanone (**1**)
$C_{cyclic}{=}O$	****	−0.00111	Cyclohexanone (**1**)
ACCHO	****	−0.00036	Benzaldeyde (**1**)
CHCOOH or CCOOH	0.08774	−0.00050	2-Methyl butanoic acid (**1**)
ACCOOH	****	0.00777	Benzoic acid (**1**)
CH_3COOCH or CH3COOC	−0.26623	0.00083	Isopropyl acetate (**1**)
$COCH_2COO$, COCHCOO or COCCOO	****	0.00036	Ethyl acetoacetate (**1**)
CO–O–CO	0.91939	0.00198	Propanoic anhydrite (**1**)
ACCOO	****	0.00001	Benzoic acid ethyl ester (**1**)
CHOH	0.03654	−0.00092	2-Butanol (**1**)
COH	0.21106	0.00175	2-Methyl-2-butanol (**1**)
$CH_m(OH)C_n(OH)$ $m,n \in (0, 2)$	****	0.00235	1,2,3-Propanotriol (**2**)
$CH_{m\ cyclic}{-}OH$ $m \in (0, 1)$	****	−0.00250	Cyclopentanol (**1**)
$CH_m(OH)CH_n(NH_p)$ $m,n,p \in (0, 2)$	****	0.00046	1-Amino-2-butanol (**1**), 1-hydroxy-*N*-methylbutylamine (**1**)

Table 2 (continued)

Groups	w_{2j}	v_{2j} (m^3 $kmol^{-1}$)	Sample assignments (occurrences)
$CH_m(NH_2)CH_n(NH_2)$ $m,n \in (0, 2)$	****	****	1,2-Propanodiamine (**1**)
$CH_{m\ cyclic}$–NH_p–$CH_{n\ cyclic}$ $m,n,p \in (0, 2)$	−0.13106	−0.00179	Pyrrolidine (**1**)
CH_m–O–CH_n=CH_p $m,n,p \in (0, 2)$	****	−0.00206	Ethyl vinyl ether (**1**)
AC–O–CH_m $m(0, 3)$	****	0.01203	Ethyl phenyl ether (**1**)
$CH_{m\ cyclic}$–S–$CH_{n\ cyclic}$ $m,n \in (0, 2)$	−0.01509	−0.00023	Tetrahydrothiophene (**1**)
CH_m=CH_n–F $m,n \in (0, 2)$	****	****	1-Fluoro-1-propene (**1**)
CH_m=CH_n–Br $m,n, \in (0, 2)$	****	−0.0058	1-Bromo-1-propene (**1**)
CH_m=CH_n–I $m,n \in (0, 2)$	****	****	1-Iodo-1-propene (**1**)
ACBr	−0.03078	0.00178	Bromotoluene (**1**)
ACI	0.00001	0.00171	Iodotoluene (**1**)
$CH_m(NH_2)$–COOH $m \in (0, 2)$	****	****	2-Aminohexanoic acid (**1**)

[a]Corrections for stress-strained effects in molecules are treated similarly to second-order groups.

Table 3
Comparison of the first- and second-order approximations of the method

X	Data points	Standard deviation		AAE		AAPE	
		1st-order	2nd-order	1st-order	2nd-order	1st-order	2nd-order
w	181	0.0291	0.0150	0.0160	0.0100	4.71%	3.00%
V_1	312	0.00236 m^3 $kmol^{-1}$	0.00192 m^3 $kmol^{-1}$	0.00139 m^3 $kmol^{-1}$	0.00105 m^3 $kmol^{-1}$	1.16%	0.89%

$$\text{Standard deviation} = \sqrt{\Sigma(X_{est} - X_{exp})^2/N}$$

$$\text{AAE} = \frac{1}{N}\sum |X_{est} - X_{exp}|$$

$$\text{AAPE} = \frac{1}{N}\sum \frac{|X_{est} - X_{exp}|}{X_{exp}}\ 100\%$$

where N is the number of experimental data, X_{est} is the estimated value of the property X, and X_{exp} is the experimental value of the property X.

and amines (C, H, O, N); (4) thioles and thialkanes (C, H, S) and (5) halogenated compounds (C, H, X). The errors in acentric factor are largest for compounds which are not "normal" in Pitzer's sense, such as halogenated compounds. However, the molar volume correlation is

Table 4
Average absolute percentage error for different types of compounds

Types of compound	Acentric factor		Liquid molar volume (m^3 $kmol^{-1}$)	
	Data points	AAPE%	Data points	AAPE%
(C,H)	114	2.5	134	0.7
(C,H,O)	36	3.6	108	0.8
(C,H,O,N)	12	4.4	27	1.4
(C,H,S)	2	1.0	8	1.5
(C,H,X)	17	4.2	35	0.4

X = halogen

Table 5
Predictions of liquid densities of amorphous polymers and comparisons with experimental data at 298 K

Polymer	Experimental value (g cm ^{3})	This method (g cm^{-3})	%Error	Elbro et al., 1991 (g cm^{-3})	%Error
Polyethylene	0.855	0.853	0.2	0.855†	0
Polypropene	0.850	0.846	0.5	0.864	−1.7
Polybutene	0.860	0.848	1.4	0.862	−0.3
Polypentene	0.850	0.849	0.1	0.861	−1.3
Polyhexene	0.860	0.850	1.2	0.861	−0.1
Polyisobutene	0.860	0.863	−0.3	0.864	−0.4
Poly(5-phenylpentene)	1.050	1.033	1.6	1.038	1.1
Polymethylacrylate	1.220	1.249	−2.4	1.205	1.2
Poly(ethylacrylate)	1.120	1.173	−4.7	1.140	−1.8
Poly(butyl methacrylate)	1.053	1.065	−1.1	1.078	−2.4
Poly(hexyl methacrylate)	1.007	1.023	−1.6	1.034	−2.7
Poly(2-ethylbutyl methacrylate)	1.040	1.009	2.9	1.037	0.3
Poly(1-methylpentyl-methacrylate)	1.013	1.021	−0.8	1.037	−2.4
Poly(octyl methacrylate)	0.971	0.997	−2.6	1.005	−3.5
Poly(dodecyl methacrylate)	0.929	0.960	−3.3	0.968	−4.2
Poly(vinyl propionate)	1.020	1.171	−14.8	1.221	−19.7
Poly(isopropyl vinyl ether)	0.924	0.917	0.7	0.925	−0.1
Poly(butyl vinyl ether)	0.927	0.947	−2.1	0.960	−3.5
Poly(*sec*-butyl vinyl ether)	0.924	0.907	1.8	0.915	1.0
Poly(isobutyl vinyl ether)*	0.930	0.931	−0.1	0.964	−3.7
Poly(pentyl vinyl ether)	0.918	0.934	−1.7	0.946	−3.0
Poly(hexyl vinyl ether)	0.925	0.925	0.0	0.935	−1.1
Poly(octyl vinyl ether)	0.914	0.911	0.3	0.920	−0.7
Poly(2-ethylhexyl vinyl ether)	0.904	0.903	0.1	0.922	−2.0
Average mean deviation			1.93%		2.42%

*Second-order groups are involved in the repeating unit
†Fitted

successful for all classes. Extrapolation to unknown compounds is an important issue in property estimation. Often a method shows good accuracy in the interpolation and regeneration of the experimental data used in the regression, but unreliable extrapolation for long-chain and complex compounds. To test the current method, we used it to predict the properties of amorphous polymers, taking values for the group contributions determined only from the low-molecular-weight compounds (Tables 1 and 2). This is in contrast to the method of Elbro et al. (1991) who directly used data on polymers (polyethylene). Here, the liquid mass density ($g\ cm^{-3}$) is predicted by adding the contributions of the functional groups of each repeating unit divided by the molecular weight of the unit (Van Krevelen and Hoftyzer, 1972). Table 5 shows the predictions of the liquid density for a variety of amorphous polymers along with their experimental values (Van Krevelen and Hoftyzer, 1972) and predictions using the method of Elbro et al., (1991) (Note that the present method is limited to 298 K while the latter method is applicable over a wide temperature range.) The results seem to be within the expected experimental uncertainty. In particular, while the method does not show subtle variations with side-chain length for substituted polyethylene, it is not clear that they exist.

5. Conclusions

The property estimation method proposed by Constantinou and Gani (1994) has been successfully applied to the estimation of the acentric factor and the liquid molar volume at 298 K. The method appears to be as accurate as the experimental data used to establish the group contributions and has a wide range of applicability. It can be used to obtain the liquid densities of a variety of amorphous polymers as well as for smaller substances.

List of symbols

A	constant in the model with values zero or unity
a, b, c	universal constants for the acentric factor
d	universal constant for the liquid molar volume
M_j	occurrences of the type j second-order group in a compound
N_i	occurrences of the type i first-order group in a compound
P_r	reduced pressure
T_r	reduced temperature
V_l	liquid molar volume
v_{1i}	contribution of the type i first-order group to the estimation of the liquid molar volume
v_{2j}	contribution of the type j second-order group to the estimation of the liquid molar volume
w	acentric factor
w_{1i}	contribution of the type i first-order group to the estimation of the acentric factor
w_{2j}	contribution of the type j second-order group to the estimation of the acentric factor

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Appendix A

To illustrate the proposed technique and demonstrate its simplicity and accuracy, we provide the estimation of the acentric factor and the liquid molar volume at 298 K for the 2,3,4-tetramethylpentane (**1**) and methylisobutyl ketone (**2**). Comparisons with experimental data are also reported (see Tables A1–A3).

Table A1
Group contribution information for the liquid molar volume

$CH_3CH(CH_3)CH(CH_3)CH(CH_3)CH_3$ (**1**)			$CH_3COCH_2CH(CH_3)CH_3$ (**2**)		
Group order	Occurrences	Contribution	Group order	Occurrences	Contribution
First			First		
CH_3	5	0.02614×5	CH_3	2	0.02614×2
CH	3	0.00711×3	CH_3CO	1	0.03655×1

Table A1 (continued)

$CH_3CH(CH_3)CH(CH_3)CH(CH_3)CH_3$ (1)			$CH_3COCH_2CH(CH_3)CH_3$ (2)		
Group order	Occurrences	Contribution	Group order	Occurrences	Contribution
			CH_2	1	0.01641×1
			CH	1	0.00711×1
		$\Sigma N_i v_{1i} = 0.15203$			$\Sigma N_j v_{1i} = 0.11235$
Second			Second		
$CH(CH_3)$ $CH(CH_3)$	2	-0.00203×2	$(CH_3)_2CH$	1	0.00133×1
			CH_3COCH_2	1	-0.00030×1
		$\Sigma M_j v_{2j} = -0.00406$			$\Sigma M_j v_{2j} = 0.00103$

Table A2
Group contribution information for the acentric factor

$CH_3CH(CH_3)CH(CH_3)CH(CH_3)CH_3$ (1)			$CH_3COCH_2CH(CH_3)CH_3$ (2)		
Group order	Occurrences	Contribution	Group order	Occurrences	Contribution
First			First		
CH_3	5	0.29602×5	CH_3	2	0.29602×2
CH	3	-0.07063×3	CH_3CO	1	1.01522×1
			CH_2	1	0.14691×1
			CH	1	-0.07063×1
		$\Sigma N_i w_{1i} = 1.26821$			$\Sigma N_i w_{1i} = 1.68354$
Second			Second		
$CH(CH_3)$ $CH(CH_3)$	2	-0.00475×2	$(CH_3)_2CH$	1	0.01740×1
			CH_3COCH_2	1	-0.20789×1
		$\Sigma M_j w_{2j} = -0.0095$			$\Sigma M_j w_{2j} = -0.19049$

Table A3
First and second-order estimations of the sample compounds (**1**) and (**2**)

Sample compound	First-order approximations	Second-order approximation	Exp. value	%Error
(**1**)	$V_1 = 0.01211 + 0.15203 = 0.16414$ m^3 kmol^{-1}	$V_1 = 0.01211 + (0.15203 - 0.00406) = 0.16008$ m^3 kmol^{-1}	0.15951 m^3 kmol^{-1}	0.4
(**2**)	$V_1 = 0.01211 + 0.11235 = 0.12446$ m^3 kmol^{-1}	$V_1 = 0.01211 + (0.11235 + 0.00103) = 0.12549$ m^3 kmol^{-1}	0.12576 m^3 kmol^{-1}	0.2
(**1**)	$w = 0.4085[\ln(1.1507 + 1.26821)]^{1/0.505} = 0.3195$	$w = 0.4085[\ln(1.1507 + 1.26821 - 0.00950)]^{1/0.505} = 0.3167$	0.3161	0.2
(**2**)	$w = 0.4085[\ln(1.1507 + 1.68354)]^{1/0.505} = 0.4430$	$w = 0.4085[\ln(1.1507 + 1.68354 - 0.19049)]^{1/0.505} = 0.3863$	0.3887	0.6